

In Silico Comparative Analysis of Genomic GC Content Across Bacterial Thermal Habitats

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Abstract:

Background:

The DNA genome of the organisms consists of bases named A, T, G and C. There is a hypothesis regarding the GC content (the percent of G and C in the genome). Since the G-C bonds are stronger than the A-T bonds, it is said that the GC is used to make the sequence stronger and more durable. Also, there is GC3, which is the GC on the third position of the protein-coding sequences. Both GC of the genome and GC3 might have a relation to this.

Results:

There are 3 thermal groups with 17 bacteria each – overall 51 bacteria. Python is used to calculate the GC content of each bacterium. The data is represented by 5 graphs. The analysis compares the GC content of bacteria across all thermal groups. The last graph examines the relation of GC to GC3 from the protein-coding sequences of each bacterium. An intriguing and unexpected dependency is found.

Conclusion:

From the results of this analysis, we can sum up that there is no defined correlation between GC content in the bacteria's genome and their thermal habitat, but a dependency between GC and GC3 in the coding sequences is found. When the GC percentage is low, the GC3 is lower, whereas when the GC is high, the GC3 is higher. Moreover, the gap between the two values is greatest at the lowest and highest points.

Background:

The DNA is composed of nucleotides, each containing a nitrogenous base, a sugar, and a phosphate group. There are four nitrogenous bases found in the DNA, classified into two groups: purines, which have a large double-ring structure, and include adenine (A) and guanine (G), and the pyrimidines, which have a smaller, single-ring structure and include cytosine (C) and thymine (T). In RNA, the thymine is replaced by another pyrimidine, called uracil (U).

The nucleobases in the DNA pair according to specific complementary base-pairing rules: adenine pairs with thymine (A-T), while guanine pairs with cytosine (G-C). An A-T pair is held together by two hydrogen bonds, whereas a G-C pair is held together by three. Consequently, G-C pairs are stronger and more stable than A-T pairs. There is a hypothesis that the number of G and C

bases in the DNA string (GC-content percentage) is determined by the environment in which the organism lives. The more G-C pairs in the DNA of the cell, the stronger the molecule is. This improves resistance to environmental changes, such as critically high temperatures that cause DNA denaturation and degradation. The GC is a huge biological factor.

The protein-coding genes in the DNA genome are composed of sequences of three-nucleotide units called codons. Each codon consists of three positions, with each position occupied by one of the four nucleobases: adenine (A), thymine (T), guanine (G) or cytosine (C). The specific sequence of bases within a codon determines which amino acid is incorporated into the resulting protein.

Although all three codon positions contribute to the genetic code, the third position is often less constrained than the first and second positions because of the degeneracy of the genetic code. Actually, the bases on the first and second position determine what the synthesized amino acid would be. In many cases, a change in the third nucleotide does not alter the encoded amino acid. This makes the third codon position particularly useful for studying patterns of nucleotide composition and potential evolutionary adaptations.

One such measure is GC3 content, which represents the percentage of codons in a protein-coding sequence in which the third nucleotide is guanine (G) or cytosine (C). Because the GC pairs have additional hydrogen atom compared with the AT pairs, a higher GC3 content may contribute to greater local DNA stability. Therefore, examining GC3 content can provide insight into the relation between nucleotide composition, genetic variation, and environmental factors.

Materials and Methods:

The following is a list of all the bacteria used in the analysis. They are sorted into 3 groups by the temperature of the environment they live in. There are 17 bacteria per each group, so overall 51 organisms' genomes are used to analyze their GC.

Psychrophiles (under 20°C):

1. *Carnobacterium maltaromaticum*
2. *Clostridium estertheticum*
3. *Colwellia psychrerythraea*
4. *Desulfotalea psychrophila*
5. *Flavobacterium psychrophilum*
6. *Marinomonas primoryensis*
7. *Moritella marina*
8. *Octadecabacter antarcticus*
9. *Octadecabacter arcticus*

10. *Planococcus halocryophilus*
11. *Polaromonas naphthalenivorans*
12. *Polaromonas vacuolata*
13. *Psychrobacter arcticus*
14. *Psychrobacter cryohalolentis*
15. *Psychroflexus torquis*
16. *Psychromonas ingrahamii*
17. *Shewanella frigidimarina*

Mesophiles (from 20°C to 40°C):

1. *Agrobacterium tumefaciens*
2. *Bacillus subtilis*
3. *Bordetella pertussis*
4. *Clostridium tetani*
5. *Corynebacterium diphtheriae*
6. *Escherichia coli*
7. *Helicobacter pylori*
8. *Lactobacillus acidophilus*
9. *Lactococcus lactis*
10. *Legionella pneumophila*
11. *Mycobacterium tuberculosis*
12. *Neisseria gonorrhoeae*
13. *Pseudomonas aeruginosa*
14. *Salmonella enterica*
15. *Staphylococcus aureus*
16. *Streptococcus pneumoniae*
17. *Vibrio cholerae*

Thermophiles (over 40°C):

1. *Alicyclobacillus acidocaldarius*
2. *Anoxybacillus flavithermus*
3. *Caldicellulosiruptor bescii*

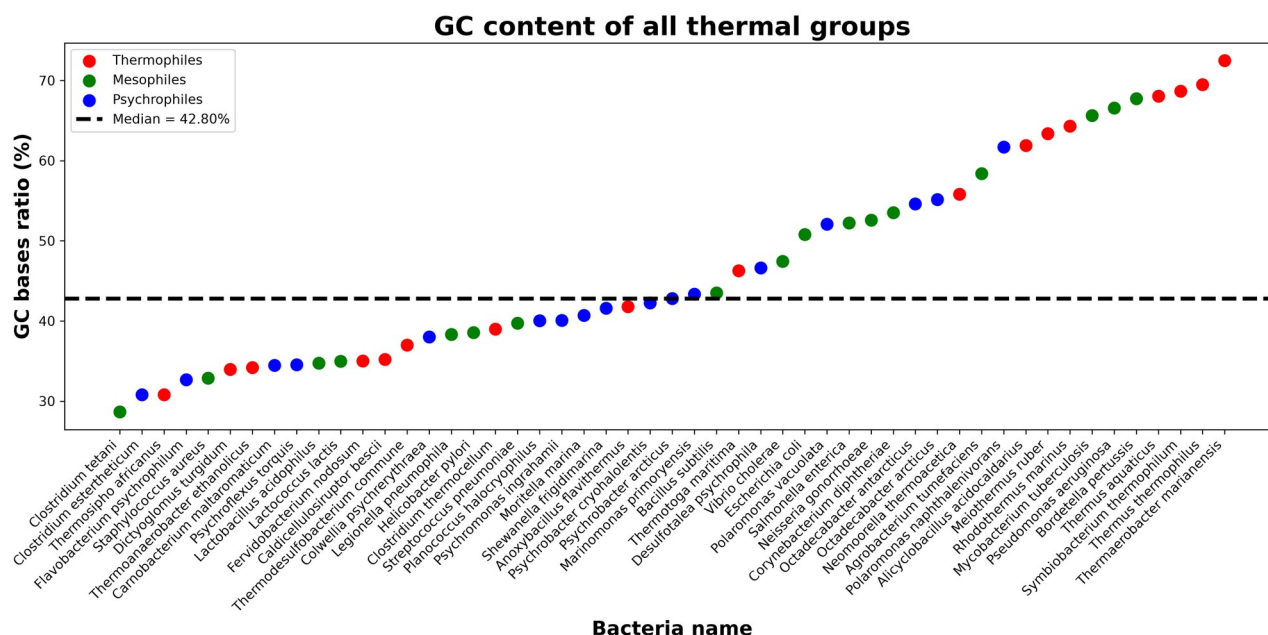
4. *Clostridium thermocellum*
5. *Dictyoglomus turgidum*
6. *Fervidobacterium nodosum*
7. *Meiothermus ruber*
8. *Neomoorella thermoacetica*
9. *Rhodothermus marinus*
10. *Symbiobacterium thermophilum*
11. *Thermaerobacter marianensis*
12. *Thermoanaerobacter ethanolicus*
13. *Thermodesulfobacterium commune*
14. *Thermosipho africanus*
15. *Thermotoga maritima*
16. *Thermus aquaticus*
17. *Thermus thermophilus*

For the calculations of the GC% in the analysis are used Python scripts with BioPython library. The graphs included in this document are created using Python scripts with Matplotlib library.

Results:

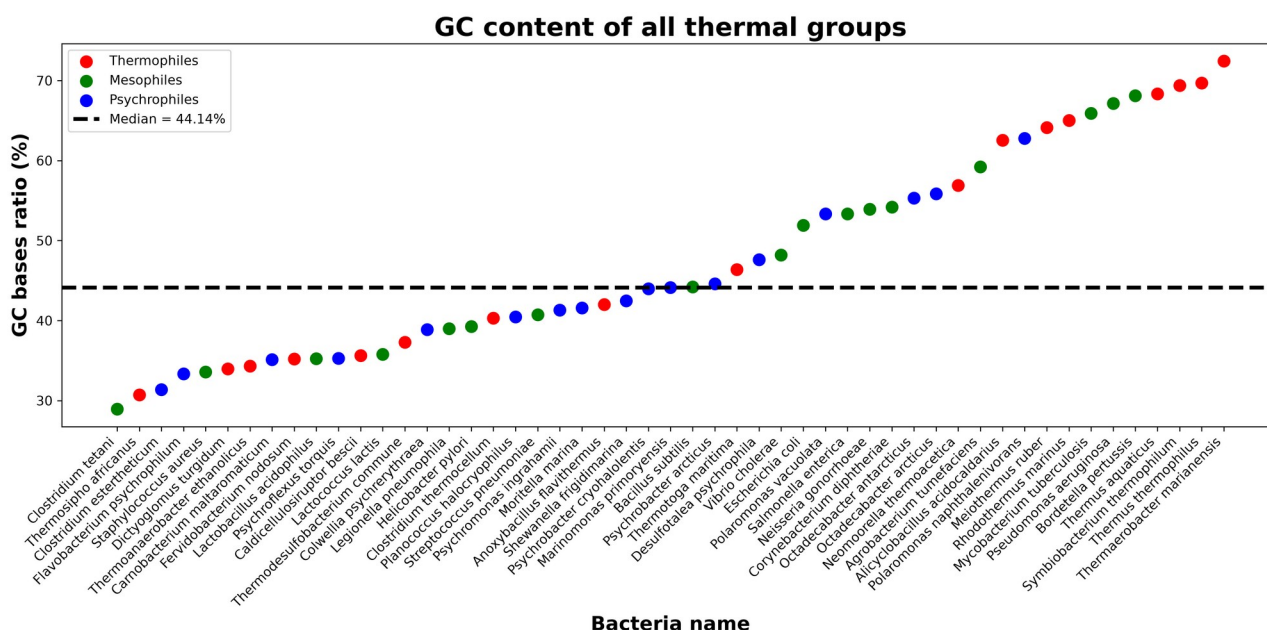
For all graphs applies that individual values are represented by colored dots according to the bacteria's thermal groups as it follows: red for Thermophiles, green for Mesophiles and blue for Psychrophiles. There is also a black dotted line, the median, showing the average of all GC values, which is useful as a reference when analyzing the distribution of the data.

Graph 1 :



The first graph shows the GC content percentage of the complete genome of each bacterium. The abscissa displays the names of all bacterial species, while the ordinate presents the corresponding GC percentage. The organisms are arranged in ascending order of genomic GC content, from the lowest to the highest value.

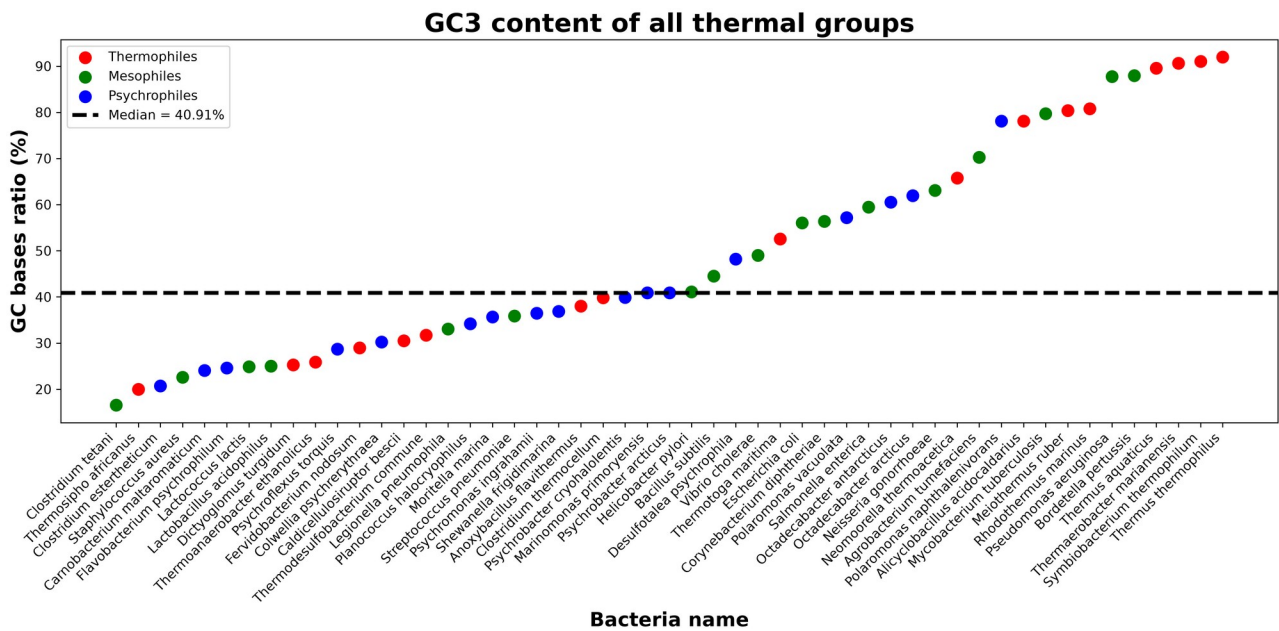
Graph 2 :



The second graph focuses on the GC content percentage of only the protein-coding genes of each bacterium. On the abscissa are displayed the names of all bacteria, whereas on the ordinate is

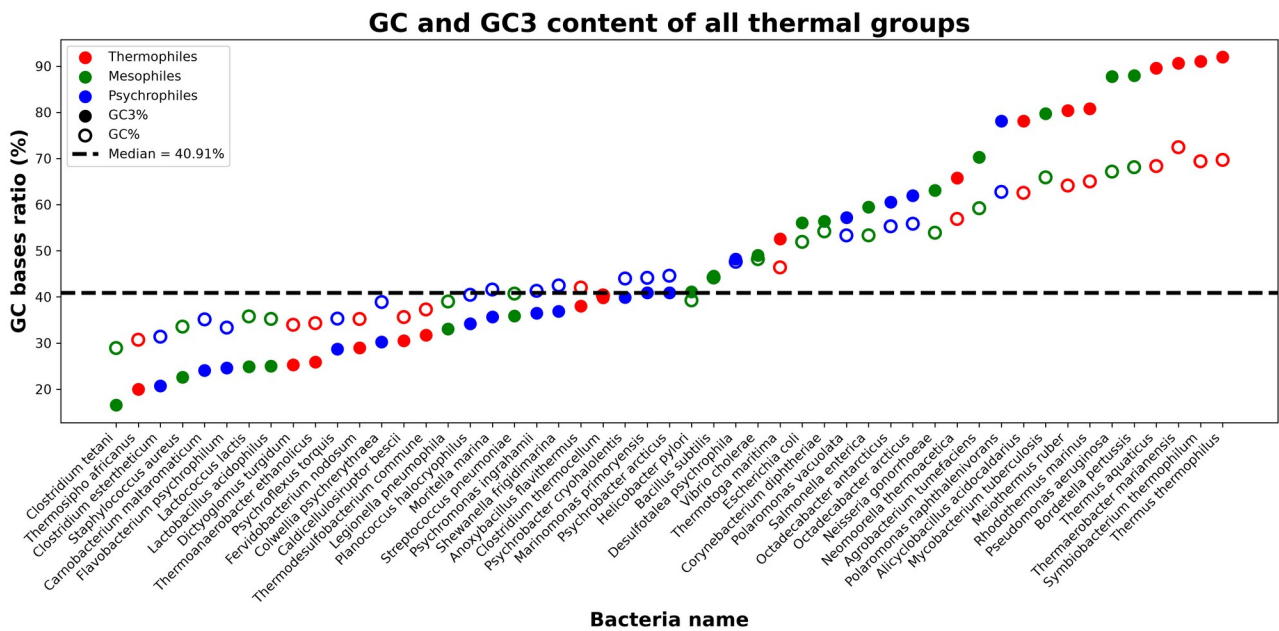
the scale for the GC percent. The bacteria are sorted accordingly to their GC percentage. More specifically, they are sorted by the GC% in their coding sequences, so that the organisms with the lowest are first and then with the highest are last.

Graph 3 :



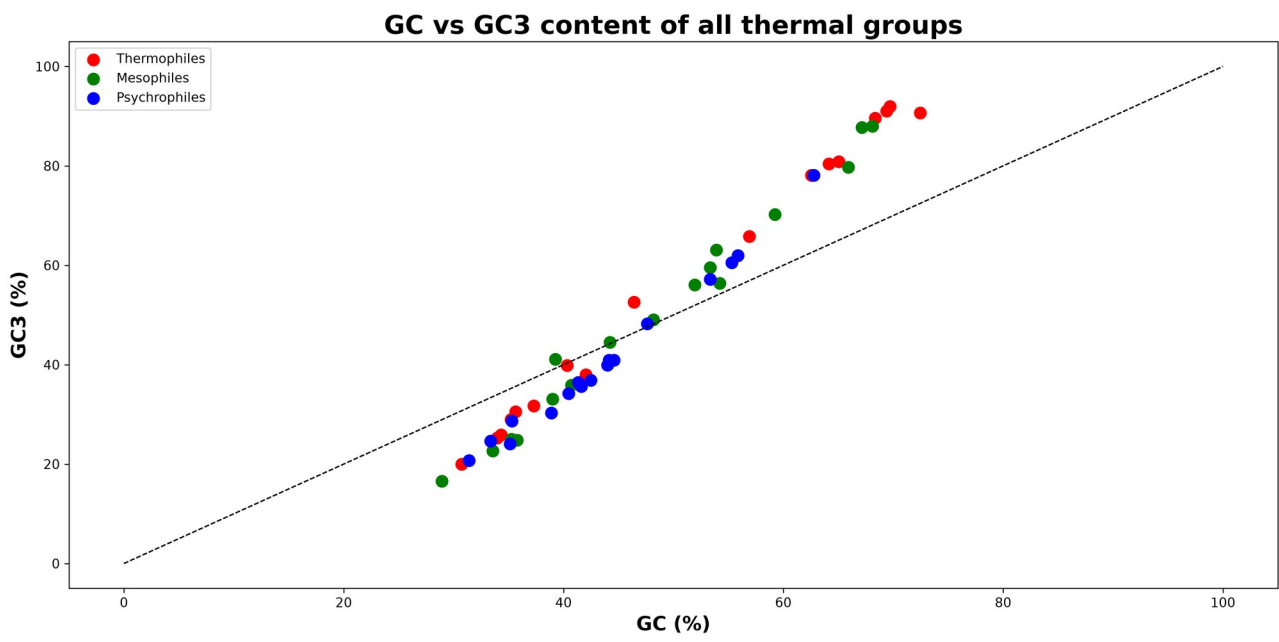
The third graph shows the GC content percentage in the third position of the codons in the protein-coding genes (GC3) of each bacterium. On the abscissa are displayed the names of all bacteria, whereas on the ordinate is the scale for GC percent. The bacteria are sorted accordingly to the GC3% in their coding sequences, from low to high.

Graph 4 :



The fourth graph shows the GC content percentage in the protein-coding genes and on the third position of the codons (GC3) of each bacterium. On the abscissa are displayed the names of all bacteria, whereas on the ordinate is the scale for the GC percent. The bacteria are sorted by the GC3% in their coding sequences, in ascending order. A distinction between the values for GC and GC3 is made by using full circles for the GC3 and hollow ones for the GC percentage. The graph actually presents the same information from Graphs 2 and 3, but on the same graph, so that it is easier to compare the values of GC and GC3 for same bacterium and therefore analyze it.

Graph 5 :



The fifth graph shows the GC content percentage in the protein-coding sequences and on the third position of the codons (GC3) of each bacterium. On the abscissa is the scale for the GC, whereas on the ordinate is the scale for the GC3 in the coding sequences. Each bacterium is represented by one dot that corresponds to its both GC and GC3 values. The bacteria are sorted by the GC3% in their coding sequences, from low to high. There is also a black dotted line called the identity line that is used as a reference to indicate where the GC and GC3 values are exactly equal. The graph focuses on the correlation between GC and GC3 to see how they differentiate from each other and if there is a dependency between them.

Discussion:

H1: Whole-genome GC content is associated with bacterial thermal habitats.

The hypothesis proposes that whole-genome GC content is related to the thermal habitat of bacteria. If such an association exists, bacteria adapted to different temperature ranges would be expected to show differences in their overall genomic nucleotide composition, potentially resulting in a separation of the thermal groups based on GC content. However, the distribution shown in Graph 1 does not reveal a separation between Psychrophiles, Mesophiles, and Thermophiles. Instead, bacteria belonging to the three groups are distributed across a broad and overlapping range of GC percentages. Some organisms from different thermal habitats have similar GC values, while bacteria within the same thermal group can also display considerable variation. Consequently, the results do not provide sufficient evidence for a relationship between whole-genome GC content and thermal habitat. The observed distribution therefore does not support the hypothesis.

H2: Coding-sequence GC content is associated with bacterial thermal habitats.

This hypothesis examines whether the GC composition of protein-coding sequences differs according to the thermal habitat of bacteria. A relation of this kind would be expected to produce a recognizable pattern in which organisms from different temperature groups occupy separate or partially distinct ranges of coding-sequence GC content. Graph 2, however, shows substantial overlap among the thermal groups, with no consistent trend linking lower or higher GC percentages to a particular temperature category. Bacteria from the same thermal group are not restricted to a specific range, and similar GC values occur among organisms adapted to different temperature conditions. This suggests that the nucleotide composition of protein-coding regions does not provide a clear distinction between the thermal habitats examined in this analysis. Therefore, the findings do not support the proposed association between coding-sequence GC content and bacterial thermal habitat, and the hypothesis is not confirmed.

H3: GC3 content is associated with bacterial thermal habitats.

The hypothesis assumes that GC3 - the GC content at the third position of codons, is associated with bacterial thermal habitats. Since the third codon position is often more variable in protein-coding genes, a relation with temperature could potentially be reflected in differences in

GC3 among the 3 thermal groups. Nevertheless, Graph 3 does not demonstrate a distinct distribution of GC3 values among them. Rather than forming separate clusters, the bacteria are widely dispersed throughout the range of observed GC3 percentages. Considerable overlap can be seen between the groups, meaning that similar GC3 values occur in bacteria belonging to different thermal categories. There is also no consistent increase or decrease in GC3 that corresponds to the temperature range associated with each group. These observations indicate that GC3 content alone does not appear to be a reliable indicator of bacterial thermal habitat within the dataset. Thus, the results do not support this hypothesis.

H4: There is a correlation between GC and GC3 contents from the protein-coding sequences.

The fourth hypothesis investigates whether the GC content of protein-coding sequences is related to their GC3 content. Unlike the previous hypotheses, the results presented in Graph 4 reveal a clear relation between these two measures. Bacteria with lower coding-sequence GC content generally have lower GC3 values, whereas organisms with higher coding-sequence GC tend to exhibit substantially elevated GC3 values.

Graph 4 also provides a useful comparison between the two measures through the median where GC and GC3 have the same value. At lower GC percentages, most data points are located below this line, indicating that GC3 is lower than the overall GC content of the coding sequences. In contrast, at higher GC percentages, the points increasingly occur above the line, showing that GC3 exceeds the corresponding overall GC value. The position of the points therefore demonstrate that the difference between GC and GC3 is not constant across the dataset.

A similar pattern can be observed in Graph 5, which focuses more directly on the difference between coding-sequence GC and GC3. Such dissimilarity becomes more noticeable in sequences with either very low or very high GC content. AT-rich coding regions, under the identity line, tend GC3 to fall substantially below the total coding-sequence GC content. On the other side of the identity line, GC-rich sequences show the opposite pattern, with GC3 becoming considerably higher than the overall GC value. This suggests that nucleotide bias at the third codon position becomes increasingly pronounced as coding sequences become strongly AT-rich or GC-rich.

The observed correlation can also be explained by the characteristics of the genetic code. The first and second codon positions are generally more constrained because substitutions at these positions are more likely to alter the encoded amino acid. In comparison, the third position is often subject to weaker amino-acid-level constraints, as many changes at this position are synonymous and do not affect the resulting protein. As a result, broader nucleotide-composition biases can be reflected particularly strongly at the third codon position. In GC-rich coding regions, this may lead to increased representation of G and C on the third position, whereas AT-rich regions may show a corresponding enrichment of A and T.

Given that, the results provide evidence in favor of the hypothesis. The strong correspondence between coding-sequence GC and GC3 demonstrates that the third codon position is closely associated with the overall nucleotide composition of protein-coding regions. At the same time, the pattern is not simply a constant one-to-one relationship: the difference between GC and GC3 becomes more evident toward the AT-rich and GC-rich extremes. This indicates that GC3

reflects the broader compositional bias of protein-coding sequences while showing greater variation at the extremes of the GC range. Therefore, the hypothesis is supported by the observed data, with Graphs 4 and 5 providing complementary evidence for the dependency between overall coding-sequence GC content and GC3 nucleotide composition at the third codon position.

Conclusion:

For this analysis, a total of 51 bacteria are used, divided into three thermal groups according to the optimal temperature of their respective habitats. Then the whole-genome sequences and protein-coding sequences of these bacteria are used to calculate their GC content using a Python program with the BioPython library. The resulting data is subsequently visualized in graphs created with Python and the Matplotlib library, which allows to examine the distribution of GC content and identify potential relations between the variables.

The analysis shows no clear correlation between thermal habitat and either whole-genome GC content or protein-coding GC3 content. The bacteria from the different thermal groups are distributed across overlapping ranges, suggesting that temperature alone does not have a strong influence on either of these measures. However, a noticeable dependency is observed of GC3 content to overall GC content in protein-coding sequences. As the GC content moves further away from the identity line, the difference between GC and GC3 generally becomes larger. In addition, bacteria with lower GC content tend to have even lower GC3 values, while those with higher GC content generally show higher GC3 values.

Overall, the results indicate that the GC content of both whole genomes and protein-coding sequences is not strongly associated with the temperature of the bacterial habitat within the dataset examined. Since GC3 may contribute to the stability and other properties of bacterial genomes, its variation may instead be influenced by other biological or evolutionary factors specific to each bacterium. Nevertheless, the observed dependency of coding-sequence GC3 to GC demonstrates that the nucleotide composition of the third codon position is closely linked to the broader GC composition of protein-coding regions.